

Ayudantía 7- MAT1620

1. Determine si los siguientes límites existen.

(a) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 4y^2}{x^2 + 2y^2}.$

(b) $\lim_{(x,y) \rightarrow (1,0)} \frac{xy - y}{(x - 1)^2 + y^2}.$

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}.$

(d) $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2).$

2. Determine el conjunto donde la función

$$f(x, y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2} & \text{si } (x, y) \neq (0, 0) \\ 1 & \text{si } (x, y) = (0, 0) \end{cases}$$

es continua.

3. Sea

$$f(x, y) = \begin{cases} \frac{x^3 y - xy^3}{x^2 + y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

Encuentre, en caso que existan, $f_x(0, 0)$ y $f_y(0, 0)$.

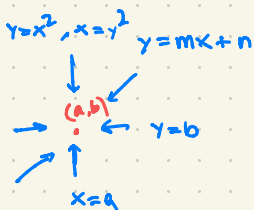
4. Calcule las primeras derivadas parciales de las siguientes funciones:

a) $f(x, y) = y^5 - 3xy.$

b) $f(x, y) = \frac{x}{y}.$

c) $f(x, y) = x^2 y e^{3xy}.$

d) $f(x, y, z) = x^2 y \cos(z/x).$



1. Determine si los siguientes límites existen.

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 4y^2}{x^2 + 2y^2}$$

$$\begin{aligned} x^2 + y^2 &= r^2 \\ x &= r \cos(\theta) \\ y &= r \sin(\theta) \\ x^2 + y^2 &= r^2 \\ \frac{y}{x} &= \tan(\theta) \end{aligned}$$

a) $x=0$:

$$\lim_{y \rightarrow 0} \frac{-4y^2}{2y^2} = -2$$

$y=0$:

$$\lim_{x \rightarrow 0} \frac{x^4}{x^2} = \lim_{x \rightarrow 0} x^2 = 0$$

el límite no existe

$$(b) \lim_{(x,y) \rightarrow (1,0)} \frac{xy - y}{(x-1)^2 + y^2}$$

$x=1$:

$$\lim_{y \rightarrow 0} \frac{y - y}{y^2} = 0$$

$y=0$:

$$\lim_{x \rightarrow 1} \frac{0}{(x-1)^2} = 0$$

$y=mx+n \rightarrow 0 = m+n \Rightarrow n = -m$

$\Rightarrow$

$$y = mx - m = m(x-1)$$

$$\lim_{x \rightarrow 1} \frac{xm(x-1) - m(x-1)}{(x-1)^2 + m^2(x-1)^2} = \lim_{x \rightarrow 1} \frac{(x-1)(xm - m)}{(x-1)^2(1+m^2)}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}^2 \cdot m}{\cancel{(x-1)}^2(1+m^2)} = \frac{m}{1+m^2} \rightarrow \text{límite no existe}$$

$$(c) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}.$$

↳ $x = 0$:

$$\lim_{y \rightarrow 0} \frac{0}{\sqrt{y^2}} = 0$$

↳ $y = mx$:

$$\lim_{x \rightarrow 0} \frac{xmx}{\sqrt{x^2 + m^2x^2}} = \lim_{x \rightarrow 0} \frac{mx^2}{\sqrt{x^2(m+1)}} = \lim_{x \rightarrow 0} \frac{mx^2}{|x|\sqrt{m+1}} = 0$$

↳ $y = x^2$:

$$\lim_{x \rightarrow 0} \frac{x^3}{\sqrt{x^2 + x^4}} = \lim_{x \rightarrow 0} \frac{x^3}{|x|\sqrt{1+x^2}} \Rightarrow \begin{aligned} \lim_{x \rightarrow 0^-} \frac{x^3}{-x\sqrt{1+x^2}} &= 0 \\ \lim_{x \rightarrow 0^+} \frac{x^3}{x\sqrt{1+x^2}} &= 0 \end{aligned}$$

↳ $x = y^2$:

$$\lim_{y \rightarrow 0} \frac{y^3}{\sqrt{y^4 + y^2}} = 0$$

↳ $x = r \cos \theta$, $y = r \sin \theta \Rightarrow (x, y) \rightarrow (0, 0) \Rightarrow r \rightarrow 0$

$$\lim_{r \rightarrow 0} \frac{r^2 \sin \theta \cos \theta}{r} = \lim_{r \rightarrow 0} r \sin \theta \cos \theta = 0$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0$$

$$(d) \lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2).$$

$$x = r \cos(\theta), \quad y = r \sin(\theta) \Rightarrow x^2 + y^2 = r^2$$

$$\hookrightarrow (x,y) \rightarrow (0,0) \Rightarrow r \rightarrow 0$$

$$\lim_{r \rightarrow 0} r^2 \ln(r^2) = 0 \cdot -\infty$$

$$\hookrightarrow = \lim_{r \rightarrow 0} \frac{\ln(r^2)}{1/r^2} = \frac{-\infty}{\infty}$$

$$\stackrel{L'H}{\Rightarrow} \lim_{r \rightarrow 0} \frac{\frac{1}{r^2} \cdot 2r}{-2/r^3} = \lim_{r \rightarrow 0} \frac{2/r}{-2/r^3} = \lim_{r \rightarrow 0} -r^2 = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2) = 0$$

2. Determine el conjunto donde la función

$$f(x,y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2} & \text{si } (x,y) \neq (0,0) \\ 1 & \text{si } (x,y) = (0,0) \end{cases}$$

es continua.

$$\hookrightarrow \lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

f es continua en (a,b)

$$\hookrightarrow x=0:$$

$$\lim_{y \rightarrow 0} \frac{0}{y^2} = 0$$

{ si el límite existiera debería cumplir

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{2x^2 + y^2} = 0 \neq f(0,0)$$

f no puede ser continua en $(0,0)$

$\Rightarrow f$ es continua en $\mathbb{R}^2 - \{(0,0)\}$

3. Sea

$$f(x, y) = \begin{cases} \frac{x^3y - xy^3}{x^2 + y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

Encuentre, en caso que existan, $f_x(0, 0)$ y $f_y(0, 0)$.

↳ $x = 0$:

$$\lim_{y \rightarrow 0} \frac{0}{y^2} = 0$$

↳ $y = mx$:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^3 mx - x \cdot m^3 x^3}{x^2 + m^2 x^2} &= \lim_{x \rightarrow 0} \frac{x^4 (m - m^3)}{x^2 (1 + m^2)} \\ &= \lim_{x \rightarrow 0} x^2 \cdot \frac{m - m^3}{1 + m^2} = 0 \end{aligned}$$

↳ $y = x^2$:

$$\lim_{x \rightarrow 0} \frac{x^3 x^2 - x \cdot x^6}{x^2 + x^4} = \lim_{x \rightarrow 0} \frac{x^5 (1 - x^2)}{x^2 (1 + x^2)} = 0$$

↳ $x = y^2$:

$$\lim_{y \rightarrow 0} \frac{y^6 y - y^2 y^3}{y^6 + y^2} = 0$$

$$|x| \leq a$$

$$-a \leq x \leq a$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{x^3 y - x y^3}{x^2 + y^2} = \lim_{(x, y) \rightarrow (0, 0)} \frac{xy(x^2 - y^2)}{x^2 + y^2}$$

$$\rightarrow |x^2 - y^2| \leq x^2 + y^2$$

$$\rightarrow |x| \cdot |y| \cdot |x^2 - y^2| \leq |x| \cdot |y| \cdot (x^2 + y^2)$$

$$\rightarrow \frac{|x| \cdot |y| \cdot |x^2 - y^2|}{x^2 + y^2} \leq |x| \cdot |y|$$

$$\left| \frac{|x| \cdot |y| \cdot |x^2 - y^2|}{x^2 + y^2} \right| \leq |x| \cdot |y|$$

$$\Rightarrow -|x| \cdot |y| \leq \frac{xy(x^2 - y^2)}{x^2 + y^2} \leq |x| \cdot |y|$$

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} -|x| \cdot |y| \leq \lim_{(x, y) \rightarrow (0, 0)} \frac{xy(x^2 - y^2)}{x^2 + y^2} \leq \lim_{(x, y) \rightarrow (0, 0)} |x| \cdot |y|$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{x^3y - xy^3}{x^2 + y^2} = 0 = f(0,0)$$

$\therefore f$ es continua en $(0,0)$

$$f_x(a,b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

$$f_y(a,b) = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}$$

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{f(h,0)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{0/h^2}{h} = 0 \quad \left\{ \begin{array}{l} f_x(0,0) = 0 \end{array} \right.$$

$$f_y(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h)}{h} = \frac{0/h^2}{h} = 0 \quad \left\{ \begin{array}{l} f_y(0,0) = 0 \end{array} \right.$$

4. Calcule las primeras derivadas parciales de las siguientes funciones:

a) $f(x,y) = y^5 - 3xy.$

$$a) \quad f_x(x,y) = \frac{\partial}{\partial x} (y^5 - 3xy) = -3y$$

$$f_y(x,y) = \frac{\partial}{\partial y} (y^5 - 3xy) = \frac{\partial}{\partial y} (y^5) - \frac{\partial}{\partial y} (3xy) = 5y^4 - 3x$$

b) $f(x,y) = \frac{x}{y}.$

b)

$$f_x(x,y) = \frac{\partial}{\partial x} \left(\frac{1}{y} \cdot x \right) = \frac{1}{y}$$

$$f_y(x, y) = \frac{\partial}{\partial y} \left(\frac{x}{y} \right) = -\frac{x}{y^2}$$

$$c) f(x, y) = x^2 y e^{3xy}.$$

$$f_x(x, y) = 2xy \cdot e^{3xy} + x^2 y \cdot e^{3xy} \cdot 3y = 2xy e^{3xy} + 3x^2 y^2 e^{3xy}$$

$$f_y(x, y) = x^2 \cdot e^{3xy} + x^2 y \cdot e^{3xy} \cdot 3x = x^2 e^{3xy} + 3x^3 y e^{3xy}$$

$$d) f(x, y, z) = x^2 y \cos(z/x).$$

$$\begin{aligned} f_x(x, y, z) &= 2xy \cdot \cos(z/x) + x^2 y \cdot -\sin(z/x) \cdot -\frac{z}{x^2} \\ &= 2xy \cos(z/x) + yz \sin(z/x) \end{aligned}$$

$$f_y(x, y, z) = x^2 \cos(z/x)$$

$$f_z(x, y, z) = x^2 y \cdot -\sin(z/x) \cdot \frac{1}{x} = -xy \sin(z/x)$$